

KEERTHANA KRISHNAN

keerthanakrishnan2601@gmail.com | +1 571-283-5765 |

[LinkedIn](#)

EDUCATION

Master of Science in Electrical and Computer Engineering

Virginia Tech, Blacksburg, VA

Dec 2024

GPA: 3.4/4

(Relevant Coursework: Computer Architecture, Advanced VLSI Design, and Advanced Analog devices)

Bachelor of Engineering in Electronics and Communication

Vidyavardhaka College of Engineering, Mysore, India

Aug 2018-Jun 2022

GPA: 8.95/10

(Relevant Courses: Analog Electronics, VLSI Circuit Design and Lab, Network Protocol, Communication Systems and lab, Embedded Systems, Power electronics and Assembly Language Programming.)

PROJECTS

- **Parkinson's Disease Detection using Python Programming**
 - Performed pioneering method employing Machine Learning, Artificial Intelligence, technology for early Parkinson's Disease (PD) detection by analyzing handwritten patterns.
 - The proposed system, utilizing a Random Forest Classifier, aims to transform PD diagnosis, facilitating timely intervention and improving overall patient care and quality of life. The project's use of Python Program underscores its accessibility and adaptability.
- **Optimization of NoCs using Spare Wire Replacement with AXI4 Lite Integration**
 - Developed a spare wire replacement algorithm to enhance fault tolerance in Network-on-Chip (NoC) systems, addressing potential scaling issues and proposing solutions such as different encoding schemes or hybrid Automatic Repeat request (ARQ) techniques.
 - Integrated the algorithm with the NoC using an AXI4 Lite interface, facilitating efficient data exchange and improving system reliability.
- **IP Verification of JTAG TAP Controller**
 - Developed and executed a detailed verification plan for the JTAG TAP controller, ensuring compliance with IEEE 1149.1 (Boundary Scan Standard). This involved designing test scenarios to validate the functionality of TAP interface signals (TDI, TMS, TCK, TDO, and TRST) and ensuring proper state transitions in the TAP controller's Finite State Machine (FSM).
 - Verified the JTAG TAP controller using SystemVerilog and UVM, ensuring functional and code coverage. Debugged waveforms to identify design issues and validate shift operations.
- **Integrated Body-Biasing Technique for Power Optimization in 90nm Master-Slave Flip-Flop Design**
 - Implemented schematic and layout in Cadence using 90nm technology for a master-slave flip-flop, utilizing Verilog testbenches for comprehensive design verification.
 - Employed body-biasing technique to minimize power consumption, showcasing expertise in analog and digital circuit design optimization.
- **ASIC Implementation of Superscalar Microprocessor Using RISC-V**
 - Initiated the microprocessor design on a Virtex-7 FPGA at 50MHz allows quick prototyping, facilitating early-stage testing and validation.
 - The subsequent shift to ASIC using Innovus enhances overall performance, lowers power consumption, and optimizes area utilization by converting FPGA logic elements into dedicated physical gates.

TECHNICAL SKILLS & INTERESTS

- **Interests:** VLSI design, CMOS Circuit Design, Digital Design, Scripting Languages, Linux Kernel Programming, Logic design, Assertion Based Verification, RISC processor, SoC Design, RTL design, Debugging, RTL design, SystemVerilog, Code coverage, Regression Testing, Formal Verification, UVM, Memory Controllers.
- Logic synthesis, Digital physical design, and implementation(P&R): Library setup, Floor plan, power planning, placement and route, clock tree synthesis, Power grid architecture routing and analysis, DRC clean up, LVS checks, timing checks and power optimizations, clock/power distribution, layout design, static timing analysis, Cross-domain clocking analysis, Physical Verification.
- **EDA Tools:** QuestaSim, 8051, Git, Virtuoso, Xilinx Vivado, Innovus, Genus, Synopsys VC and Spice.
- **Misc Software:** Xilinx ISE, Keil µVision, MATLAB, Pycharm, Riverbed, Simulink and Scilab.
- **Programming Languages:** Python, C/C++, System Verilog, TCL and Perl.
- **Operating Systems:** Windows, Linux.
- **Soft Skills:** Effective communication, Analytical Skills, Self-motivated, Team collaboration, Communication Skills, Adaptable and Flexible, Problem-Solving Skills and Interpersonal Skills.

AWARDS & ACHIEVEMENTS

- Best Outgoing Student and Semi-Finalist in Intra-college IEEE Bot event.